



Application of γ -aminobutyric acid (GABA) improves fruit quality and rootstock drought tolerance in apple

Pengda Cheng^a, Qianyu Yue^a, Yutian Zhang^a, Shuang Zhao^a, Abid Khan^c, Xinyue Yang^a, Jieqiang He^a, Shicong Wang^a, Wenyun Shen^a, Qian Qian^a, Wanshan Du^a, Fengwang Ma^a, Dehui Zhang^{a,b,**}, Qingmei Guan^{a,*}

^a State Key Laboratory of Crop Stress Biology for Arid Areas/Shaanxi Key Laboratory of Apple, College of Horticulture, Northwest A&F University, Yangling, Shaanxi, 712100, China

^b State Key Laboratory of Crop Stress Biology for Arid Areas/Shaanxi Key Laboratory of Apple, College of Life Science, Northwest A&F University, Yangling, Shaanxi, 712100, China

^c Department of Horticulture, The University of Haripur, Haripur, 22620, Pakistan

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ABSTRACT

GABA (γ -aminobutyric acid) plays a multifaceted role in plant growth, fruit quality, and tolerance to abiotic stresses. However, its physiological roles and mechanisms in the fruit quality and response to long-term drought stress in apple remain unelucidated. To investigate the effect of GABA on apple fruit quality and drought tolerance, we sprayed exogenous GABA on apple cultivar “Cripps Pink” and irrigated rootstock M.9-T337 with GABA, respectively. Results showed that exogenous GABA could effectively improve the fruit quality of “Cripps Pink”, including increased sugar-to-acid ratio, flesh firmness, pericarp malleability, and GABA content, as well as reduced fruit acidity. In addition, pretreatment of M.9-T337 plants with GABA improved their tolerance to both long- and short-term drought stress. Specifically, 1 mM exogenous GABA increased the net photosynthetic rate, relative leaf water content, root-to-shoot ratio, and water use efficiency under long-term drought stress, and delayed the increased of the relative electrolyte leakage under short-term drought stress. RNA-seq analysis identified 1271 differentially expressed genes (DEGs) between nontreated and GABA-pretreated plants under short-term drought stress. Kyoto Encyclopedia of Genes and Genomes (KEGG) analysis of these DEGs revealed that GABA may enhance plant drought resistance by upregulating the expression of genes related to “Biosynthesis of secondary metabolites”, “MAPK signaling pathway”, “Glutathione metabolism”, and “Carbon fixation in photosynthetic organisms”. In conclusion, these results revealed that exogenous GABA can improve fruit quality and enhance drought tolerance in apple.

1. Introduction

Apple is one of the most popular fruits grown worldwide with its high nutritional value and diverse flavors (Vallée Marcotte et al., 2022). The Loess Plateau is one of the most productive apple-growing regions in the world. However, drought stress occurs regularly in this area, which is a major limiting factor for apple quality and productivity (Peng et al., 2017). Drought tolerance in apples has recently been improved using a variety of methods, including application of exogenous chemical substances. Chemical compounds can alter plants' memory or their own

natural response that plants do not naturally utilize as an emergent mechanism (Merewitz et al., 2016). In addition, exogenous chemical compounds, which have higher reliability and stability, are considered safe and cost-effective, allowing the human nutrition and growing demands of the human population to be met (Merewitz et al., 2016). For example, the application of abscisic acid (ABA) to blueberries has been shown to improve their firmness, implying a ripening delay effect (Buran et al., 2012). Under drought stress, exogenous ABA promotes the lipid and flavonoid metabolism of tea plants (Gai et al., 2020). Methyl jasmonate (MeJA) is a volatile hormone that affects the quality and

* Corresponding author.

** Corresponding author. State Key Laboratory of Crop Stress Biology for Arid Areas/Shaanxi Key Laboratory of Apple, College of Horticulture, Northwest A&F University, Yangling, Shaanxi, 712100, China.

E-mail addresses: tsu.zhangdehui@163.com (D. Zhang), qguan@nwfau.edu.cn (Q. Guan).

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preservation of harvested fruit (Wang et al., 2021a). In addition, exogenous application of 5-aminolevulinic acid also improves tomato fruit color and quality by regulating carotenoid metabolism (Wang et al., 2021b). Dopamine treatment improves flavonoid and soluble sugar contents, nutrient uptake fluxes, and drought tolerance in apples (Gao et al., 2018; Liang et al., 2018; Wang et al., 2021c). Furthermore, exogenous melatonin maintains the quality of harvested apples by mediating sucrose metabolism (Fan et al., 2022).

γ -Aminobutyric acid (GABA) was first discovered in potato tubers, and subsequently found in a wide range of plants, animals, and microorganisms (Dhakal et al., 2012). GABA is a nonproteinogenic four-carbon amino acid that is mostly generated via GABA shunt, in which glutamate is decarboxylated to produce GABA by glutamate decarboxylase (GAD). GABA is also derived from polyamines, such as spermidine and putrescine, under the catalysis of diamine oxidase and polyamine oxidase. GABA is catabolized to succinic semialdehyde and then either succinate or γ -hydroxybutyrate (Shelp et al., 2012). Studies have demonstrated that GABA is an important component in the synthesis of primary and secondary metabolites (Bown and Shelp, 2016), and participates in various physiological and biochemical processes in plants, such as nitrogen and carbon metabolism, regulation of cell membrane pH (Ramesh et al., 2017), osmotic potential (Vijayakumari and Puthur, 2016), redox status (Jin et al., 2019), pollen tube growth (Priya et al., 2019), and adventitious root development (Xie et al., 2020).

Besides its role in the synthesis of primary and secondary metabolites, GABA also acts as a signaling molecule in plants (Bown and Shelp, 2016). Plants exposed to environmental stresses exhibit a rapid increase in their GABA content (Shelp et al., 2012; Bown and Shelp, 2016; Srivastava et al., 2021). Numerous studies have demonstrated that the application of exogenous GABA has a positive role in plant the mitigation of abiotic stress, including drought (Abdel Razik et al., 2021; Xu et al., 2021), salinity (Khanna et al., 2021; Wu et al., 2020), temperature (Asgarian et al., 2022; Li et al., 2016; Nayyar et al., 2014), and heavy metal (Seifikalhor et al., 2020; Wang et al., 2021d; Zhao et al., 2020) stresses. The effects of GABA on postharvest fruit quality in apples have been widely reported. For example, during postharvest storage, exogenous GABA improves apple fruit quality by regulating malate metabolism, ethylene anabolism, polyamine metabolism, and the GABA shunt pathway (Li et al., 2021a). In addition, exogenous GABA treatment significantly alleviates chilling injury of peach, zucchini, cucumber, and grapefruit during storage (Asgarian et al., 2022; Malekzadeh et al., 2017; Palma et al., 2019; Shang et al., 2011). In pears, exogenous GABA application delays peel browning by maintaining mitochondrial structure and increasing the efficacy of the mitochondrial oxidative defense mechanism (Li et al., 2019a). Similarly, exogenous GABA has also shown to prolong the postharvest life of citrus fruits by regulating citrate metabolism and amino acid accumulation (Sheng et al., 2017). Recently it was discovered that 0.5 mM GABA improves the short-term drought resistance of *Malus hupehensis* (Liu et al., 2021). However, the effect of exogenous GABA on fruit quality during fruit growth and long-term drought tolerance is mainly unknown.

In the current study, we evaluated the effects of exogenous GABA on fruit quality of “Cripps Pink” during fruit development and tolerance of dwarf rootstock M.9-T337 to drought stress. Results showed that exogenous GABA can improve fruit quality of “Cripps Pink” through increasing sugar-to-acid ratio, flesh firmness, pericarp malleability, and GABA content, as well as reducing fruit acidity. In addition, exogenous GABA increased the net photosynthetic rate, relative leaf water content, root-to-shoot ratio, and water use efficiency under long-term drought stress, and delayed the increased of the relative electrolyte leakage under short-term drought stress.

2. Materials and methods

2.1. Plant materials and treatment

“Cripps Pink” grafted onto M26 rootstock was cultivated in Yangling City, Shaanxi Province, China, while GABA was purchased from Shanghai Sangon Biotech Company. For fruit treatment with GABA, 12 healthy Cripps Pink fruit trees were randomly selected. For fruit GABA treatment, the treatment lasted for 140 days from 20 days after flowering to fruit ripening. During the treatment, fruits were sprayed with 0, 5, 10, and 50 mM GABA (with 0.05% Tween-20) eight times with one spray every 20 days. During the treatment, some of the fruit bags were randomly removed as non-bagged control. Each treatment has three biological replicates. After ripening, 20 fruits per treatment were harvested for quality determination.

For short-term drought treatment, tissue-cultured apple dwarf rootstock M.9-T337 was rooted and transplanted to a greenhouse for additional 2 months. M.9-T337 plants were then irrigated with GABA solution with various concentrations (0, 0.5, 1, 5, and 10 mM) for 30 days before short-term drought treatment (20 days). To detect the soil moisture content, the soil volumetric water content (VWC) was measured by TDR (FS6430, Spectrum Technologies, Aurora, USA) while mass water content (MWC) was measured by a weighing method. $MWC = (\text{soil weight} - \text{soil dry weight}) / (\text{soil saturated weight} - \text{soil dry weight})$. Then M.9-T337 plants were treated with short-term drought for 20 days until VWC and MWC reached 5% and 20%, respectively, followed by leaf collection for the relative electrolyte leakage assays.

For long-term drought treatment, M.9-T337 plants were pretreated with various concentrations of GABA (0, 0.5, 1, 5, and 10 mM) for 30 days before long-term drought treatment (60 days). VWC and MWC of the well-watered plants (the control group) was 43–48% and 70–100%, respectively. For the long-term drought treatment, the plants were irrigated daily to maintain 18–23% of VWC and 45–55% of MWC.

2.2. Determination of fruit quality

For external quality determination, 20 fruits with similar size were randomly selected from each treatment to measure fruit weight, shape, and skin color parameters. An electronic Vernier caliper was used to measure maximum longitudinal and transverse diameters, and ratio of longitudinal and transverse diameters were used to determine the fruit shape. Measurement of fruit skin color parameters was conducted by a portable colorimeter (Cr-100, X-Rite, USA) that was calibrated with standard white plate. Skin color changes were expressed using lightness (L^*), red-greenness (a^*), and yellow-blueness (b^*) values. Fruit yield was determined by weighing all fruits harvested from each treatment.

For internal quality determination, the soluble solid content (SSC) was determined with a PAL-1 digital refractometer (Atago, Japan). Fruit acidity was measured with a digital GMK-835F (G-WON, Korea). Flesh firmness, pericarp firmness, flesh compactness, pericarp malleability, and flesh brittleness were measured at five directions along the equatorial surface of the fruit with a fruit-texture analyzer (TMS-Touch, FTC, USA), and their average value was calculated.

For the amino acid measurement, fruit flesh was ground into fine power, which was then resuspended in the extraction solution (50% ethanol). The free amino acids were extracted with ultrasonic treatment for 20 min at 50°C. After centrifugation, the supernatant was filtered by 0.22 μm filter membrane, and then diluted 10 times for HPLC analysis as described previously (Li et al., 2019b).

2.3. Determination of relative electrolyte leakage (REL)

After 20 days of short-term drought treatment, fresh leaves were collected and washed with deionized water. The leaves were cut into leaf discs, then the leaf discs were transferred to glass tubes containing 8 ml deionized water for 4 h. The electrolyte leakage values before and after

boiling (30 min) were recorded by a conductivity meter (DSS-307; SPSIC, Shanghai, China). The relative electrolyte leakage was calculated as $REL (\%) = (T1 - T0)/(T2 - T0) \times 100$, where T1 was the value for the electrolyte leakage before boiling, T2 was the value after boiling, and T0 was the value of deionized water.

2.4. Determination of relative water content (RWC)

The RWC of the leaves was measured by following a previously reported method (Li et al., 2019c). The fresh leaves were weighed (W1) and transferred to deionized water for 12 h. After weight determination (W2), the leaves were transferred to a drying chamber for 30 min at 105°C, then at 65°C to a constant weight (W). The relative water content was calculated as:

$$RWC (\%) = \frac{(W1 - W)}{(W2 - W)} \times 100$$

2.5. Determination of water use efficiency (WUE)

Water use efficiency can be represented by carbon isotopic composition ($\delta^{13}C$). The $\delta^{13}C$ analysis was conducted as described previously (Wang et al., 2018). Briefly, the leaves were transferred to a drying chamber for 30 min at 105°C, then at 65°C to a constant weight. The dry leaf samples were ground into fine powder and filtered with a sieve (80 holes per cm^2). Lastly, $\delta^{13}C$ was determined using a stable carbon isotope mass spectrometer (Flash EA 1112 HT-Delta V Advantages, Thermo Fisher Scientific).

2.6. Measurement of photosynthetic parameters and SPAD

The LI-COR 6400 portable photosynthetic system (LI-COR 6400, Huntington Beach, CA, USA) was used to determine photosynthetic parameters. Determination of leaf photosynthesis was performed as described previously (Xie et al., 2021). The mature fifth to sixth leaves from the apical bud were measured between 09:00 and 11:00 a.m. in fine weather. The photosynthetic rate (P_n), stomatal conductance (G_s), and transpiration rate (T_r) of each leaf were measured using LED red and blue light source with $1,000 \mu mol m^{-2} s^{-1}$ light intensity and $500 \mu mol s^{-1}$ air flow, with a leaf temperature of $25 \pm 2^\circ C$ and relative humidity of the sample cell of $22 \pm 2\%$. SPAD (Soil and Plant Analyzer Development) measurement was taken with SPAD-502 (Konica Minolta, Japan). The mature fifth to sixth leaves from the apical bud were measured, and each leaf was measured twice.

2.7. Transcriptome analysis

For transcriptomic analysis, the M.9-T337 roots pretreated with 0 or 1 mM GABA for 30 days under control or short-term drought stress (20 days) were selected as experimental materials, and three biological replicates of each treatment were used. The total RNA was extracted as described previously (Chang et al., 1993). The library was generated with NEBNext® Ultra™ RNA Library Prep Kit (NEB, USA) and sequenced with the Illumina HiSeq platform at Novogene (Beijing, China). Clean reads were mapped to the latest release of the *Malus × domestica* genome sequence using HISAT2 v2.1.0 (GDDH13version1.1). Subsequent data post-processing including BAM conversion, sorting, and indexing was performed by SAMTools v1.9. Gene reads were detected by htseq-count in HTSeq v0.13.5, and gene annotation was performed by TAIR10. The R package DESeq2 v1.20.1 was used to analyze gene expression differences based on p -values below 0.05 and an absolute value of $\log_2$ (fold change) greater than 1 or 0.5. The g:Profiler was used for Gene Ontology (GO) enrichment analysis and Kyoto Encyclopedia of Genes and Genomes (KEGG) analysis. Online bioinformatics site (<http://www.bioinformatics.com.cn/>) was used to plot Venn diagrams and heat maps.

2.8. Statistical analysis

Statistical analysis was carried out by Student's two-tailed t -test. Asterisks indicate significant differences (* $p < 0.05$, ** $p < 0.01$, and *** $p < 0.001$).

3. Results

3.1. Long-term exogenous GABA application does not affect external traits of apple fruits

In general, fruits have a broad set of external and internal traits. The external fruit quality includes color, shape, size, and lack of defects. The internal quality encompasses taste, texture, aroma, nutritional value, sweetness, acidity, shelf life, and lack of defects (Musacchi and Serra, 2018). To understand the effects of exogenous GABA application on fruit internal and external quality, we sprayed GABA with various concentrations on 12-year-old "Cripps Pink". The external fruit quality (color, shape, and size) was not affected by GABA sprays, except for a slightly higher brightness values in non-bagged fruits as well as a lower red-greenness values in bagged fruits at 50 mM GABA concentration. In the bagged fruit treatment, the fruit weight in the group receiving under 5 mM treatment was lower than that of the control group. (Fig. S1).

3.2. Long-term exogenous GABA application affects the internal qualities of apple fruits

We next investigated the effect of exogenous GABA application on the internal qualities of the fruits in apple during fruit development. Soluble solid content (SSC) and acidity are two of the fundamental internal fruit traits, as well as important factors in determining the fruit flavor. After long-term application of exogenous application of GABA, we measured SSC and acidity of bagged and non-bagged fruits treated with different GABA (0, 5, 10, and 50 mM) concentrations. The results suggested that there was no significant change in SSC of the fruits after application of exogenous GABA (Fig. 1A). However, acidity of the bagged fruits decreased by 20.4%, 15.7%, and 18.9%, while that of non-bagged fruits decreased by 32.4%, 21.6%, and 29.1% under exogenous GABA (5, 10, and 50 mM) treatments, respectively, as compared with the control (0 mM GABA) group (Fig. 1B). In addition, as compared with the control, the sugar-to-acid ratio increased by 24.9%, 21.4%, and 19.3% in bagged fruits and 50.9%, 20.3%, and 36.9% in non-bagged fruits under different concentrations (5, 10, and 50 mM) exogenous GABA application, respectively (Fig. 1C). Thus, our results suggested that long-term application of exogenous GABA reduced the fruit acidity and improved the sugar-to-acid ratio during fruit development.

Since the texture parameters of fruit can be perceived by touch, whether the product is picked up by hand or chewed, we next examined the flesh firmness of both bagged and non-bagged apples when treated with various concentrations of GABA. Results showed that the flesh firmness of all the tested fruits increased significantly after application of exogenous GABA. Besides, compared with the 0 mM GABA-treated group, the application of exogenous GABA (5, 10, and 50 mM) increased flesh firmness by 11.7%, 5.5%, and 4.8% in bagged fruits and 6.4%, 3.9%, and 4.6% in non-bagged fruits, respectively (Fig. 2A). Exogenous GABA (5, 10, and 50 mM) increased pericarp malleability by 1.2%, 3.0%, and 4.6% in bagged fruits and 1.0%, 3.8%, and 2.9% in non-bagged fruits, respectively (Fig. 2B). However, our data showed no significant differences in flesh brittleness, pericarp firmness, or flesh compactness (Fig. 2C–E).

3.3. Exogenous GABA treatment can increase fruit GABA and Glu content

Glutamate (Glu) is a precursor of GABA synthesis (Li et al., 2021b). We therefore determined whether exogenous GABA treatment affected

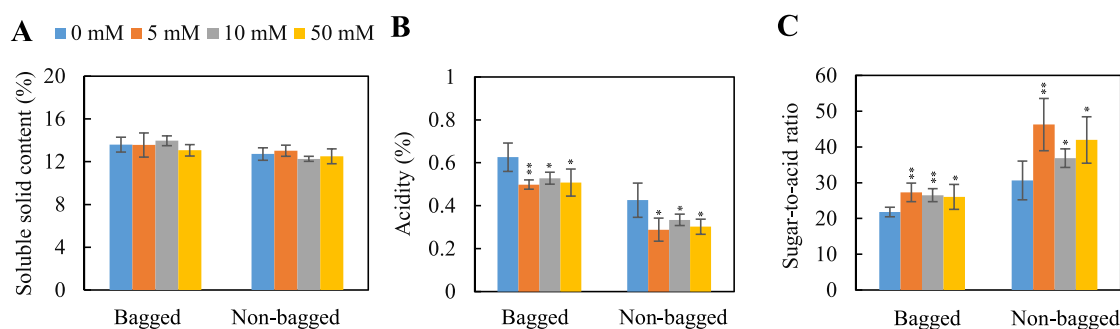


Fig. 1. Effect of various concentrations (0, 5, 10, and 50 mM) of GABA on soluble solid content (SSC), acidity, and sugar-to-acid ratio of “Cripps Pink” fruits (A–C). Each fruit was sampled from five directions. (A) The soluble solid content of the fruit, (B) the acidity of the fruit, and (C) the sugar-to-acid ratio of the fruit. Asterisks indicate the significant differences (Student’s two-tailed *t*-test; **p* < 0.05 and ***p* < 0.01). Error bars indicate SD (*n* = 5).

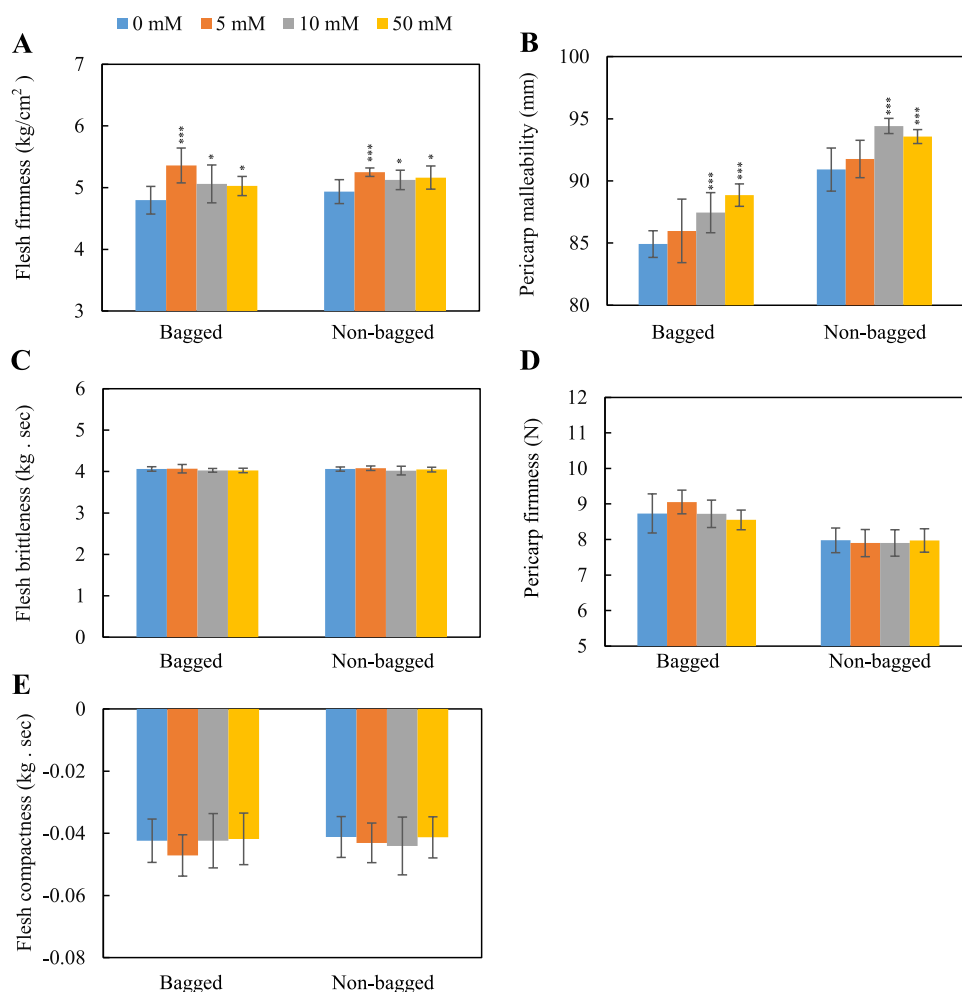


Fig. 2. Effect of various concentrations (0, 5, 10, and 50 mM) of GABA on the internal qualities of “Cripps Pink” fruits (A–E). Each fruit was measured at five directions. (A) The flesh firmness of the fruit, (B) the pericarp malleability of the fruit, (C) the flesh brittleness of the fruit, (D) the pericarp firmness of the fruit, and (E) the flesh compactness of the fruit. Asterisks indicate the significant differences (Student’s two-tailed *t*-test; **p* < 0.05 and ****p* < 0.001). Error bars indicate SD (*n* = 10).

the changes of fruit GABA and Glu content during development. As shown in Fig. 3, the long-term exogenous GABA application increased endogenous GABA and Glu concentration in both bagged and non-bagged fruits. The GABA concentration was increased by 22.7%, 28.0%, and 22.5% in bagged fruits and 16.3%, 21.8%, and 54.6% in non-bagged fruits in response to exogenous GABA (5, 10, and 50 mM) application (Fig. 3A). The Glu concentration was increased by 29.5%

and 25.7% in bagged fruits and 32.4% and 79.7% in non-bagged fruits in response to exogenous GABA (10 and 50 mM) application (Fig. 3B).

3.4. Exogenous GABA increased tolerance of apple rootstock M.9-T337 to short-term and prolonged drought stress

M.9-T337 is a widely used dwarf rootstock of apple trees; however, it

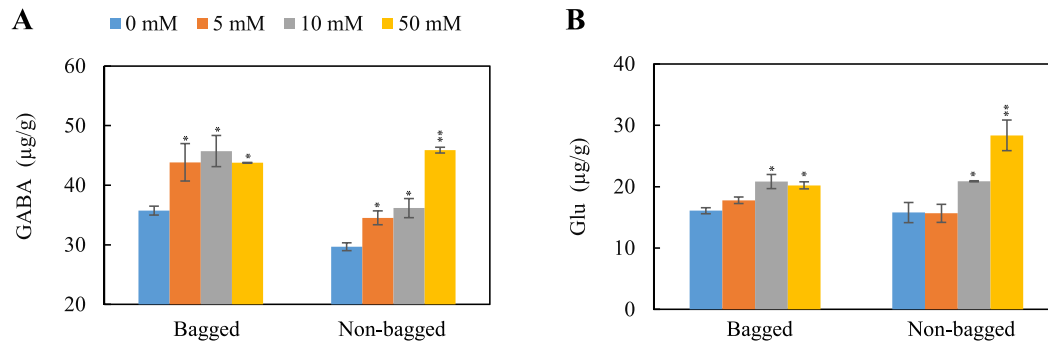


Fig. 3. Effect of various concentrations (0, 5, 10, and 50 mM) of GABA on GABA and Glu content of “Cripps Pink” fruits (A, B). Fruit flesh was used for the determination of amino acid content, GABA (γ -aminobutyric acid), and Glu (glutamate). (A) The GABA content of the “Cripps Pink” and (B) the Glu content of the “Cripps Pink”. Asterisks indicate the significant differences (Student’s two-tailed *t*-test; **p* < 0.05 and ***p* < 0.01). Error bars indicate SD (*n* = 3).

is vulnerable to drought stress because of its shallow and weak root system. To understand whether exogenous GABA benefits drought tolerance of M.9-T337, we pretreated the M.9-T337 plants with different concentrations of GABA before short-term drought stress. After 20 days of short-term drought treatment, the leaf wilting degree of exogenous GABA-treated (0.5 and 1 mM) M.9-T337 was lower than that of control plants (0 mM GABA) (Fig. 4A). Consistently, the relative electrolyte leakage (REL) of M.9-T337 pretreated with 0.5 and 1 mM GABA was lower than that of M.9-T337 treated with other GABA concentrations. In addition, the high GABA concentrations (5 and 10 mM) did not alleviate the adverse effect of drought stress on the M.9-T337 plants as determined by the leaf wilting degree and REL assays (Fig. 4B). These results revealed that 0.5 and 1 mM GABA was able to improve the short-term drought tolerance of M.9-T337.

Given that the apple plants in the Loess Plateau are frequently exposed to long-term moderate drought stress, we next investigated the effects of exogenous GABA on the dwarf rootstock M.9-T337 in response to long-term moderate drought. Under control conditions, there was no difference in plant height, root length, shoot dry weight, root dry weight, and root-to-shoot ratio among various GABA pretreatment, except that 10 mM GABA pretreatment inhibited plant height of M.9-T337 (Fig. 5A). Compared with the control group, the long-term drought stress inhibited the growth of M.9-T337. However, 0.5 and 1 mM exogenous GABA pretreatment substantially alleviated this

inhibition. The exogenous GABA (0.5 and 1 mM) promoted M.9-T337 to accumulate more biomass and have higher root-to-shoot ratio, greater height and root length under long-term drought stress, as compared with other GABA pretreatment concentrations (0, 5, and 10 mM) (Fig. 5B–F), suggesting that GABA pretreatment (0.5 and 1 mM) improves tolerance of M.9-T337 to long-term drought stress. In contrast, the 10 mM GABA pretreatment significantly inhibited the growth of M.9-T337 under drought conditions as well as in the control group (Fig. 5B–E). In addition, 5 mM GABA pretreatment had no significant effect on M.9-T337 growth under either well-watered conditions or long-term drought stress (Fig. 5B–E).

3.5. Exogenous GABA enhanced the photosynthetic capacity of M.9-T337 under long-term drought stress

We also measured photosynthetic parameters of GABA pretreated M.9-T337 under control and long-term drought conditions. Under control conditions, the GABA-pretreated M.9-T337 showed no significant difference in the photosynthetic rate (*P_n*), stomatal conductance (*G_s*), or transpiration rate (*T_r*) compared with nontreated plants. After long-term drought conditions, *P_n*, *G_s*, and *T_r* declined significantly in all the tested plants, but these parameters were obviously higher in 0.5 and 1 mM GABA-pretreated M.9-T337 than in the untreated plants. Furthermore, high concentrations of GABA (5 and 10 mM) exhibited no positive or

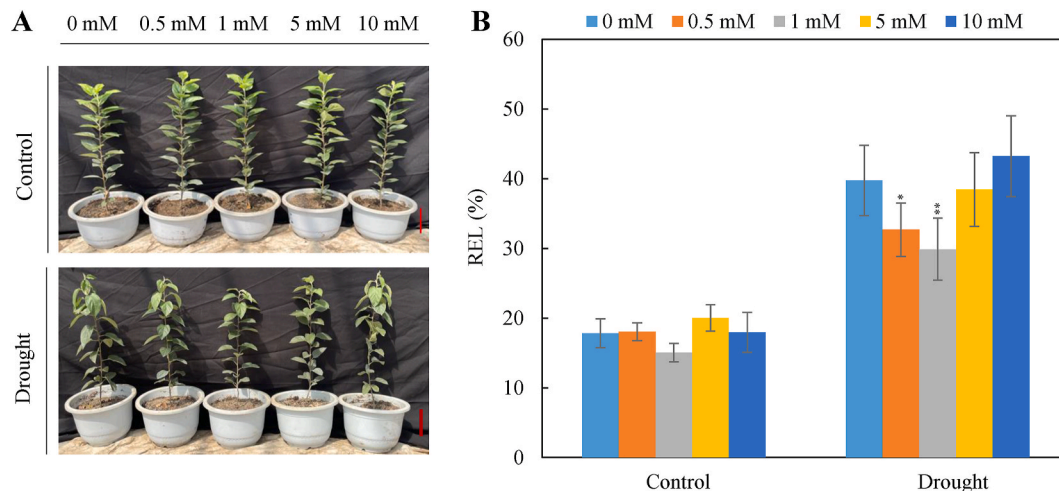


Fig. 4. Short-term drought response of M.9-T337 irrigated with various concentrations (0, 0.5, 1, 5, and 10 mM) of GABA (A, B). The degree of leaf wilting and leaf relative electrolyte leakage (REL) of M.9-T337 were determined after 20 days of short-term drought treatment. (A) Morphology of GABA-pretreated M.9-T337 plants in response to short-term drought stress. (B) REL of M.9-T337 plants irrigated with different concentrations of GABA under well-watered conditions and short-term drought stress. Data are the mean $\pm$ SD (*n* = 6). The red line is 10 cm scale. Asterisks indicate the significant differences (Student’s two-tailed *t*-test; **p* < 0.05 and ***p* < 0.01).

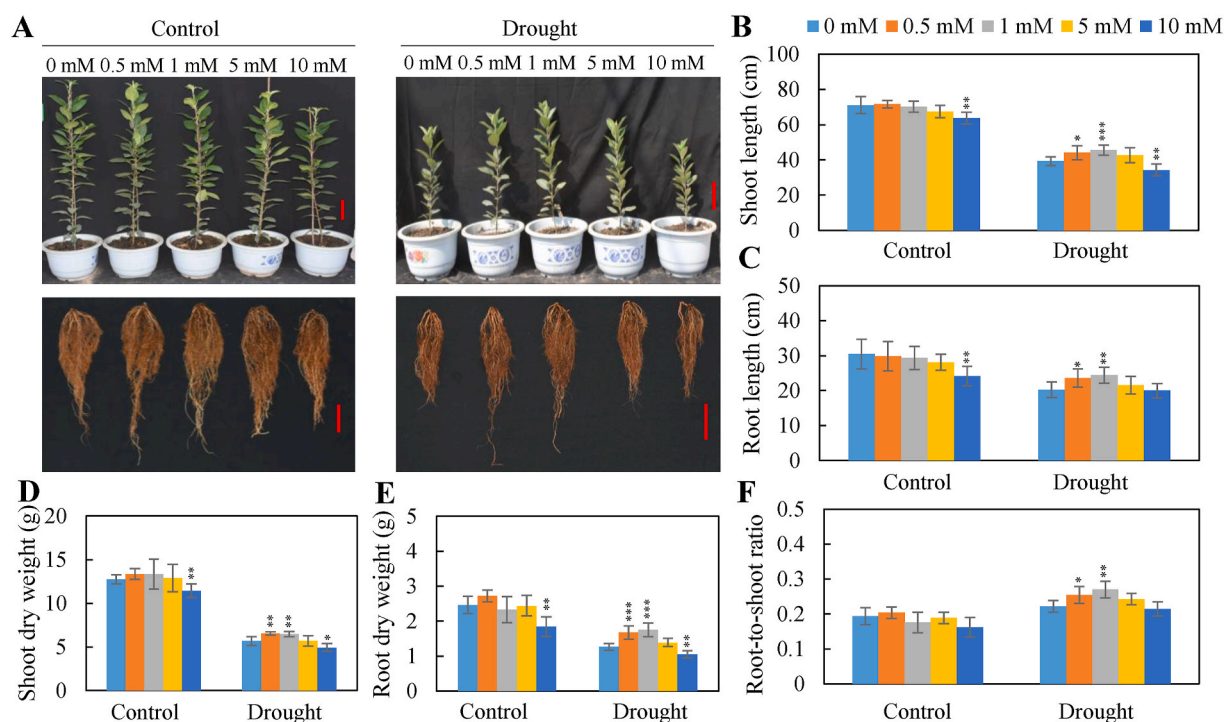


Fig. 5. Long-term drought response of M.9-T337 plants irrigated with different concentrations (0, 0.5, 1, 5, and 10 mM) of GABA (A–F). Biomass of M.9-T337 plants was determined after 60 days of long-term drought treatment. (A) The morphology of M.9-T337 plants, (B) the shoot length of M.9-T337 plants, (C) the root length of M.9-T337 plants, (D) dry weight of the M.9-T337 plants shoot, (E) dry weight of M.9-T337 plants root, and (F) the root-to-shoot ratio of M.9-T337 plants pretreated with various concentrations of GABA under well-watered conditions and long-term drought stress. The red line is 10 cm scale. Asterisks indicate the significant differences (Student's two-tailed *t*-test; **p* < 0.05, ***p* < 0.01, and ****p* < 0.001). Error bars indicate SD [*n* = 8 (B, C) and 6 (D, E, F)].

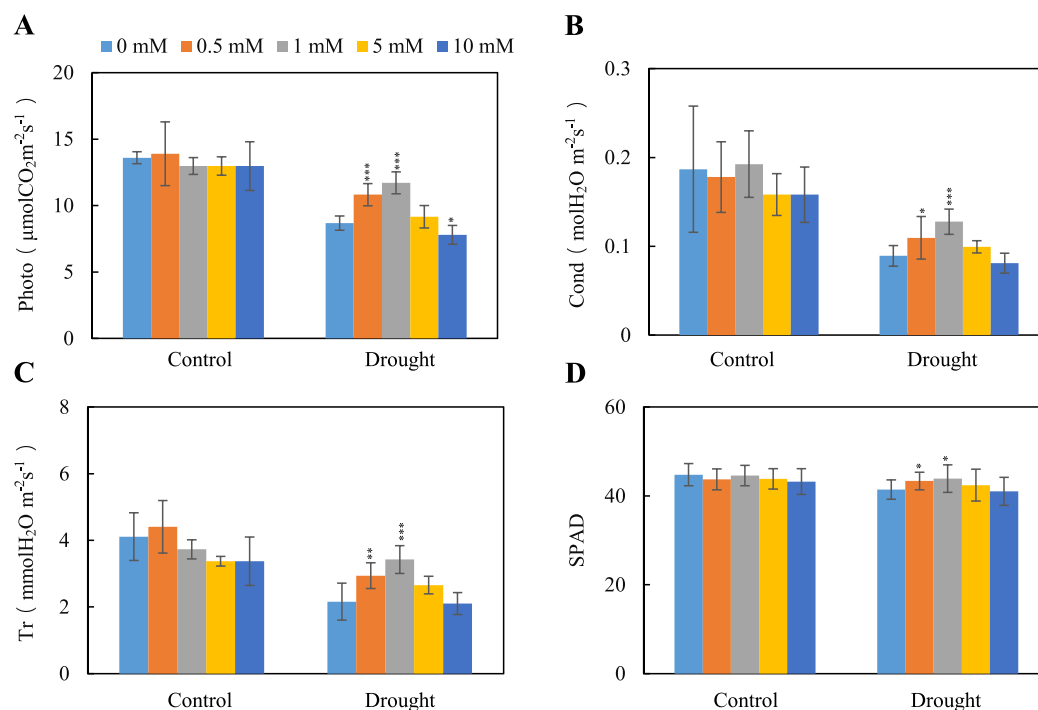


Fig. 6. Photosynthetic parameters and SPAD of the M.9-T337 plants pretreated with different concentrations (0, 0.5, 1, 5, and 10 mM) of GABA under well-watered conditions and long-term drought stress (A–D). (A) The photosynthetic rate of the M.9-T337 plants, (B) the stomatal conductance of the M.9-T337 plants, (C) the transpiration rate of the M.9-T337 plants, and (D) the SPAD (Soil and Plant Analyzer Development) of the M.9-T337 plants. Asterisks indicate the significant differences (Student's two-tailed *t*-test; **p* < 0.05, ***p* < 0.01, and ****p* < 0.001). Error bars indicate SD [*n* = 5 (A, B, C) and 29 (D)].

negative effect on photosynthesis (Fig. 6A–C). Additionally, the SPAD analysis indicated that the total chlorophyll content in GABA (0.5 and 1 mM)-pretreated M.9-T337 was higher than that in nontreated plants (Fig. 6D). These results further suggested that 0.5 and 1 mM exogenous GABA application exhibits a positive effect on the drought tolerance of M.9-T337.

3.6. Exogenous GABA increased M.9-T337 water holding and usage under long-term drought stress

Increasing relative water content (RWC) and improving water use efficiency (WUE) are critical ways in which plants adapt to drought stress. To further investigate the effect of exogenous GABA on long-term drought tolerance of M.9-T337, we examined the RWC and WUE of M.9-T337 under control and long-term drought conditions. Under long-term drought conditions, 0.5 and 1 mM GABA pretreatment increased RWC by 3.0% and 3.6%, respectively, as compared with the control (0 mM GABA) (Fig. 7A). On the contrary, no significant difference of RWC was observed among other GABA pretreatment concentrations (0, 5, and 10 mM) under long-term drought stress.

Since WUE is positively correlated with $\delta^{13}\text{C}$ (Wang et al., 2018), we used carbon isotopic composition ($\delta^{13}\text{C}$) to evaluate the whole-plant WUE of apple trees. The leaf samples pretreated with 0, 1, and 10 mM GABA were collected for the $\delta^{13}\text{C}$ determination. As shown in Fig. 7B, the $\delta^{13}\text{C}$ values were comparable in GABA pretreated and nontreated M.9-T337 under control conditions. However, 1 mM GABA pretreatment increased the $\delta^{13}\text{C}$ value of M.9-T337 under long-term drought conditions, as compared with pretreatment of 0 and 10 mM GABA, suggesting that exogenous GABA improves WUE of M.9-T337 under long-term drought conditions (Fig. 7B).

The above results suggested that pretreatment of GABA increases water holding and usage of M.9-T337 under long-term drought stress, and 1 mM GABA is enough to improve RWC and WUE of M.9-T337 under prolonged drought stress.

3.7. Exogenous 1 mM GABA promoted GABA shunt in M.9-T337

Previous studies have revealed that exogenous GABA and stress treatment can promote GABA shunt (Li et al., 2019b; Serraj et al., 1998). Therefore, we examined expression levels of the genes related to GABA synthesis and metabolism after 1 mM GABA pretreatment. As shown in Fig. 8, the pretreatment of 1 mM GABA could upregulate the expression of *MdGAD1*, *MdGAD2*, *MdGAD3*, *MdGAD4*, and *MdSSADH*. Moreover, the drought stress could upregulate the expression of *MdGAD1*, *MdGAD4*, and *MdSSADH*. However, the expression of *MdGABA-T1* and *MdGABA-T2* did not change significantly under 1 mM GABA pretreatment and drought conditions. The results indicated that 1 mM

exogenous GABA and drought stress could promote GABA shunt in M.9-T337.

3.8. Transcriptomic profiling of 1 mM GABA-pretreated plants under drought stress

To gain molecular insights into the roles of exogenous GABA on drought tolerance in apple rootstock, we performed a transcriptomic analysis of the roots pretreated with 0 and 1 mM exogenous GABA under control and drought conditions (Tables S4 and S5). Pearson correlation coefficient analysis showed a good correlation between the biological replicates (Fig. S3). The RNA-seq result revealed a total of 1271 GABA responsive genes under drought stress; among these, 299 genes were upregulated and 972 genes were downregulated by exogenous GABA (Fig. 9A and Table S5). GO enrichment analysis revealed that 1271 differential expressed genes (DEGs) were enriched in terms involved in the biological process, such as response to stimulus, stress, and hormones (Fig. 9B Table S2). Venn diagram analysis also revealed that 74 DEGs were upregulated and 320 DEGs were downregulated by both exogenous GABA application and drought stress (Fig. 9C and D). KEGG analysis of these up- and downregulated genes showed that upregulated genes were enriched mainly in the biosynthesis of secondary metabolites, MAPK signaling pathway, glutathione metabolism, and carbon fixation in photosynthetic organisms, while downregulated genes were enriched mainly in nitrogen metabolism and amino acid metabolism (Fig. S4 and Tables S6 and 7).

To understand the effects of 1 mM GABA pretreatment on M.9-T337, we compared the transcriptome data between GABA pretreated and untreated (0 mM GABA) M.9-T337 under well-watered conditions. The RNA-seq results revealed a total of 1312 GABA-responsive genes; among these, 708 genes were upregulated and 604 genes were downregulated by exogenous GABA (Fig. S5A). GO enrichment analysis revealed that 708 upregulated genes were enriched in terms involved in the biological process, such as response to, JA, SA, ABA, amino acid, organic acid, and chemical stimulus (Fig. S5B).

Next, we selected seven genes to confirm the reliability of the RNA-seq results with RT-qPCR. Homologs of *NCED6*, *ERF1*, *RD26*, *LEA*, *GSTU8*, and *ABI1* are positive regulators of drought stress (Cheng et al., 2013; Fujita et al., 2004; Iuchi et al., 2001; Liang et al., 2019; Liu et al., 2009; Zhang et al., 2021), while homolog of *CYP707A3* was a negative regulator (Umezawa et al., 2006). As shown in Fig. 10, the expression of *MdNCED6*, *MdERF1*, *MdRD26*, *MdLEA*, *MdGSTU8*, and *MdABI1* was higher in GABA-treated plants than that in non-treated plants, whereas *MdCYP707A3* was inhibited by exogenous GABA applications (Fig. 10), which was consistent with the RNA-seq data. These data also suggested that increased tolerance of M.9-T337 by GABA pretreatment might be attributed to altered expression of these drought-responsive genes.

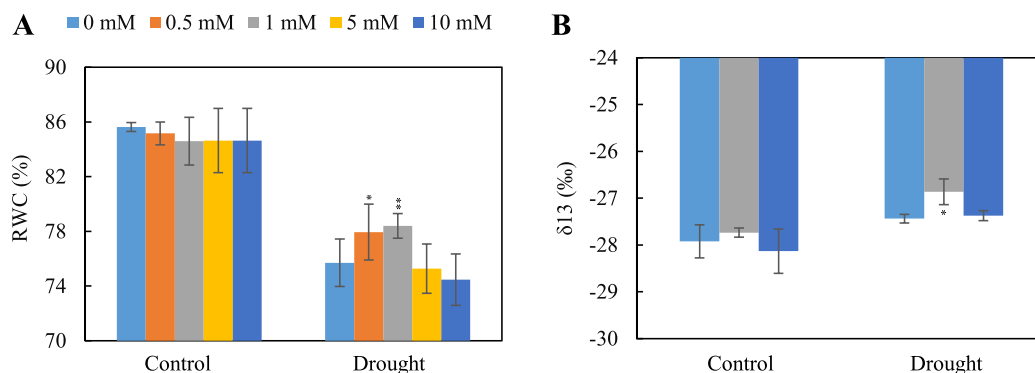


Fig. 7. The relative water content (RWC) and the water use efficiency (WUE) of M.9-T337 plants pretreated with different concentrations (0, 0.5, 1, 5, and 10 mM) of GABA under well-watered conditions and long-term drought stress (A, B). (A) The RWC of the M.9-T337 plants, (B) the WUE of the M.9-T337 plants. Asterisks indicate the significant differences (Student's two-tailed *t*-test; **p* < 0.05 and ***p* < 0.01). Error bars indicate SD [*n* = 8 (A) and 3 (B)].

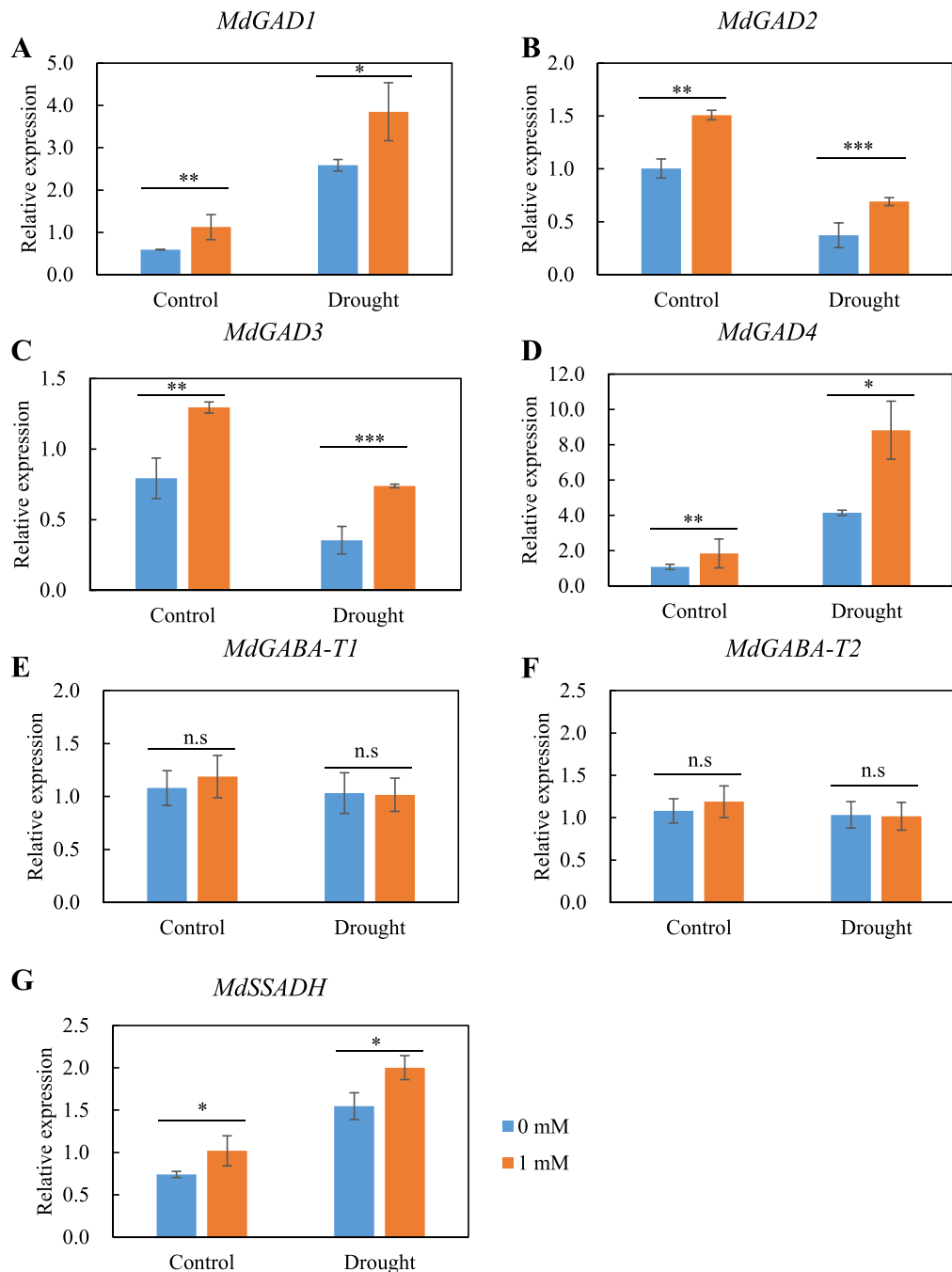


Fig. 8. Expression of genes related to GABA metabolism (A–G). M.9-T337 plants pretreated with GABA for 30 days were treated with drought for 20 days, and the roots were collected for total RNA extraction. Asterisks indicate the significant differences (Student's two-tailed t -test; * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$). Error bars indicate SD ($n = 3$).

4. Discussion

So far, our knowledge of the effects of pre- and postharvest GABA treatment have been limited to fruit physiology during storage. Preharvest GABA sprays inhibit development of soft scald with no effect on other physiological processes during apple fruit storage (Al Shoffe et al., 2021). In “Cripps Pink” fruits, the postharvest GABA treatment enhances succinate and oxalate levels while delaying the loss of titratable acidity and malate content (Han et al., 2018). It has been demonstrated that GABA postharvest dips trigger the GABA shunt and polyamine metabolism in “Golden Delicious” fruits, thus preventing ethylene anabolism (Li et al., 2021a). In current study, we characterized the effects of

preharvest GABA sprays on the harvest indices of apple fruits. Specifically, we found that flesh firmness and pericarp malleable were significantly increased, but the pericarp firmness, flesh compactness, and brittleness were not influenced remarkably (Fig. 2). Furthermore, compared with the control group, the GABA treatment dramatically reduced acidity, resulting in a higher sugar-to-acid ratio at harvest (Fig. 1). Moreover, we discovered that GABA-treated fruits had higher levels of endogenous GABA and Glu (Fig. 3). Hence, it is possible that exogenous GABA spraying could have influenced the GABA metabolism.

Despite the fact that consumers judge a product by its appearance, they eventually decide whether to buy it again on the basis of its edible quality (Vanoli and Bucchini, 2012). If the initial firmness values at

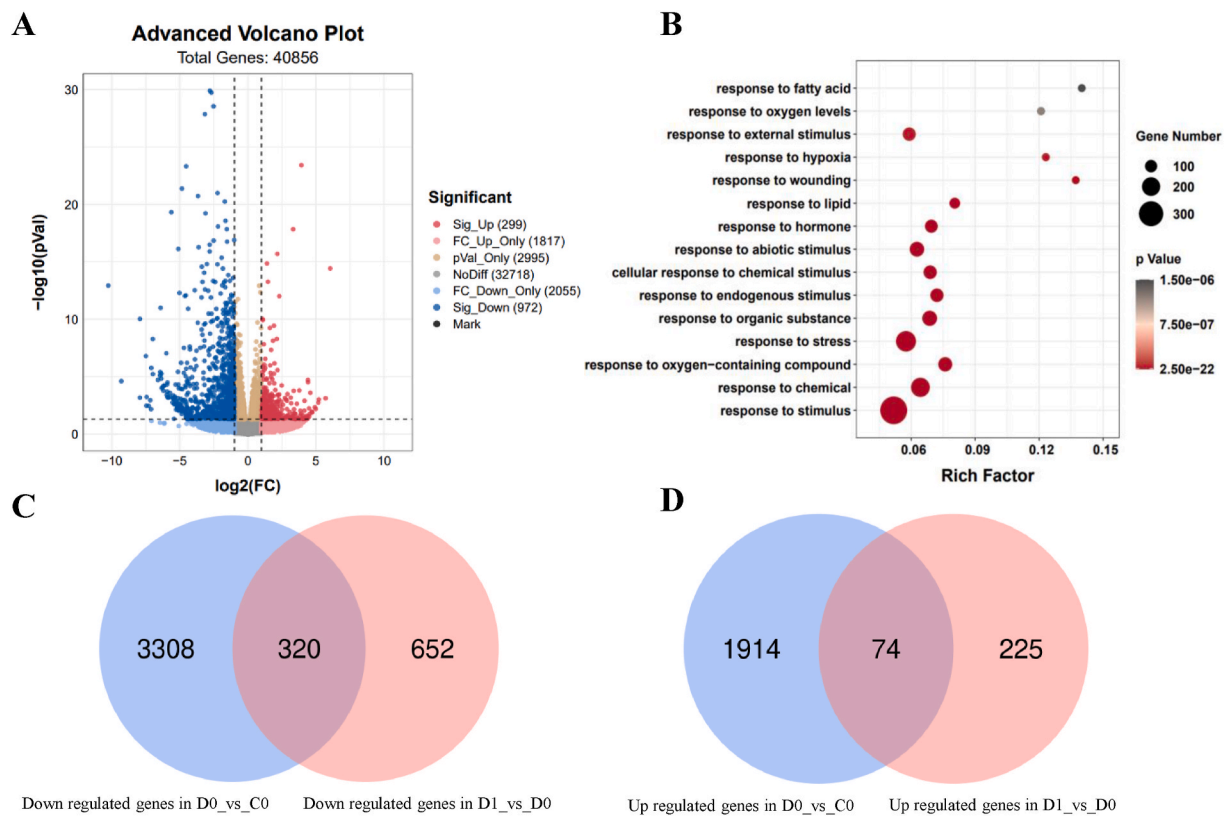


Fig. 9. Transcriptome profiling of GABA-pretreated plants under drought stress (A–D). (A) Volcano map of differentially expressed genes (DEGs) in transcriptome. (B) Gene Ontology (GO) enrichment analysis of D1_vs_D0 DEGs. (C) Venn diagram analysis showing the downregulated DEGs of D0_vs_C0 and D1_vs_D0. (D) Venn diagram analysis showing the upregulated DEGs of D0_vs_C0 and D1_vs_D0. C0, 0 mM GABA-treated plants under well-watered conditions; D0, 0 mM GABA-treated plants under drought stress; D1, 1 mM GABA-treated plants under drought stress. D0_vs_C0, differentially expressed genes under 0 mM GABA treatment under drought and well-watered conditions. D1_vs_D0, differentially expressed genes under 0 mM GABA and 1 mM GABA treatments and drought conditions.

harvest are high, or if the flesh softening rates during storage are low, or both, then the flesh firmness retention can be extended (Wu et al., 2021). According to previous research, consumers prefer apples with greater firmness such as “Honeycrisp”, which is cultivated for the fresh market (Srisantharajah et al., 2021). The use of foliar calcium sources has been shown to improve the flesh firmness of the sweet cherries (Ekinici et al., 2016). Despite the fact that there are few investigations on pericarp malleability, it is strongly linked to postharvest storage quality and is an important factor for long-term storage of fruits. However, the effects of exogenous GABA applications on flesh firmness and pericarp malleability, whether at pre- or postharvest stage, are mostly unexplored. In this study, the apple fruit sprayed with exogenous GABA exhibited greater flesh firmness as well as better pericarp malleability (Fig. 2), suggesting that preharvest GABA applications have a positive effect on postharvest fruit storage or consumer purchase behavior.

Internal fruit quality is determined by acidity and sweetness, which determines the taste and flavor of the fruit (Musacchi and Serra, 2018). Citrate and malate are the most predominant organic acid in citrus and apple fruits, respectively. During storage, the application of exogenous GABA has reduced the consumption of citrate caused by respiration and energy metabolism (Sheng et al., 2017). Despite the fact that a previous study found that 10 mM GABA treatment greatly delays the loss of titratable acidity in postharvest fruits by accelerating malate biosynthesis and inhibiting respiratory metabolism (Han et al., 2018), we observed its opposite effects, that preharvest GABA application reduced the acidity in apple fruit at harvest (Fig. 1B). Consistent with our results, a previous study reported that postharvest GABA treatment distinctly suppressed respiratory rate, reduced titratable acidity, and maintained a higher sugar-to-acid ratio during storage (Li et al., 2021a). In addition, preharvest GABA sprays on trees slightly inhibited titratable acidity in

the fruit at harvest (Al Shoffe et al., 2021). Given that citrate accumulation could be driven by malate accumulation in vacuolar (Etienne et al., 2013), the lower acidity in preharvest GABA treatment fruits in our study may be a consequence of the impaired capacity to store malate, but not the respiratory consumption.

Currently we noticed that the preharvest GABA application accelerated the accumulation of endogenous GABA. More interestingly, glutamate, the precursor of GABA, was also accumulated in response to GABA sprays (Fig. 3). Consistent with our findings, it has been observed that exogenous GABA application increased the endogenous GABA and glutamate content in apple during postharvest storage (Li et al., 2021a). In citrus, it has been discovered that exogenous GABA treatment significantly increases the accumulation of glutamate due to the feedback inhibition of glutamate decarboxylase (GAD) activity. Therefore, we speculate that an increase in the glutamate content of the preharvest GABA-treated fruits might also be a consequence of the feedback inhibition of the GAD activity.

Plants being subjected to drought stress are characterized by sluggish growth and development, reduced crop yield, and even plant death (Zhu, 2016). To cope with harsh environments, plants draw a variety of materials within themselves. Under drought stress conditions, the GABA accumulates rapidly in many plants as its accumulation has been reported in soybean (Serraj et al., 1998), sesame (Bor et al., 2009), creeping bentgrass (Merewitz et al., 2012), and tomato (Pal et al., 2016). Furthermore, it has also been reported that GABA accumulates rapidly in response to drought stress (Liu et al., 2021). A previous study showed that GABA is not only an important metabolite, but also a signal molecule in plants; hence, manipulating GABA is an effective strategy to better manage plants' response to stress (Bown and Shelp, 2016). Previously, exogenous GABA treatments were limited to their effects on

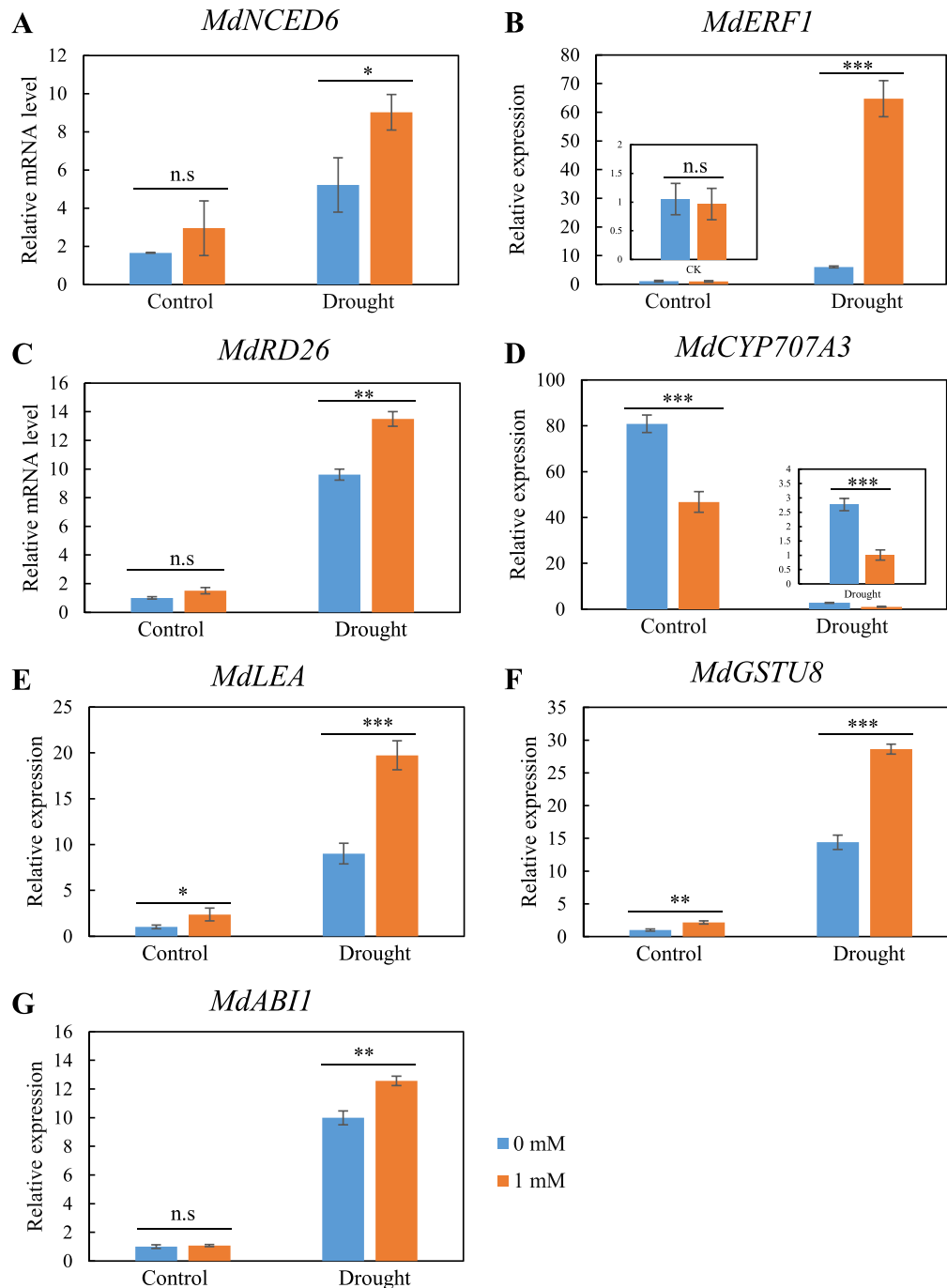


Fig. 10. Validation of drought-related genes in the transcriptomic data (A–G). M.9-T337 plants pretreated with GABA for 30 days were treated with drought for 20 days, and the roots were collected for total RNA extraction. Asterisks indicate the significant differences (Student's two-tailed *t*-test; **p* < 0.05, ***p* < 0.01, and ****p* < 0.001). Error bars indicate SD (*n* = 3).

short-term drought tolerance by spraying (Liu et al., 2021). In the current study, we characterized the effects of irrigated GABA on long-term drought tolerance in apple dwarf rootstock. We found that 0.5 mM GABA irrigation is enough to increase long-term drought tolerance of M.9-T337 (Figs. 5–7).

We have revealed evidence to support the positive effects of exogenous GABA in alleviating plant response to long-term drought stress. First, under long-term drought stress, the plants irrigated with exogenous (0.5 and 1 mM) GABA exhibited greater height, lengthier root system, and higher root-to-shoot ratio as compared with the nontreated plants (Fig. 5). Secondly, consistent with the higher biomass in GABA-

treated plants, 0.5 and 1 mM application of GABA increased the net photosynthetic rate as well as stomatal conductance and transpiration rate under drought stress, suggesting its involvement in photosynthesis by reducing stomatal limitation (Fig. 6). Thirdly, $\delta^{13}\text{C}$ measurement suggested that GABA-irrigated plants achieved a higher WUE (Fig. 7B), indicating a drought adaptation strategy to conserve water by increasing WUE (Price et al., 2002). Fourth, drought-positive regulators, including *NCED6*, *ERF1*, *RD26*, *LEA*, *GSTU8*, and *ABI1*, were induced by exogenous GABA, while the negative regulator *CYP707A3* was inhibited (Fig. 10), which might contribute to the drought response of GABA-irrigated plants. These observations suggest that exogenous GABA irrigation

alleviates plant response to long-term drought stress.

Furthermore, the upregulated DEGs induced by both drought stress and exogenous GABA in M.9-T337 were enriched in the terms of “Biosynthesis of secondary metabolites”, “MAPK signaling pathway”, “Glutathione metabolism”, and “Carbon fixation in photosynthetic organisms” (Fig. S4). Secondary metabolites such as dopamine, melatonin, and anthocyanins have various biological functions including signaling, regulating enzyme activity, and enhancing plant defense. Previous studies have found that secondary metabolites can enhance plant resistance to stress by scavenging ROS, which could protect plants from lipid peroxidation and other oxidative damage (Yadav et al., 2021). The MAPK cascade pathway consists of MAPK, MAPK kinase (MAPKK), and MAPK kinase kinase (MAPKKK), and plays a critical role in regulating plant growth, development, and stress responses. In rice, MAPK kinase has been found to enhance drought tolerance by phosphorylating and activating the MPK6 and MPK3, respectively (Ma et al., 2017). In Arabidopsis, it was noted that MAPKKK18 specifically phosphorylates MAPKK3, which activates downstream MAPK1/2/7/14, thereby improving plant drought tolerance (Li et al., 2017). Additionally, glutathione is widely present in plants and animals and has antioxidant effects and integrates detoxification. It has been found that glutathione can enhance plant tolerance to drought by scavenging ROS, maintaining cellular redox homeostasis, and regulating protein function (Ji et al., 2018). Finally, the upregulated genes were enriched in “Carbon fixation in photosynthetic organisms”, which is also consistent with the elevated photosynthesis and biomass of the plants treated with GABA (0.5 and 1 mM). Taken together, GABA may enhance the M.9-T337 drought tolerance through activating genes related to the secondary metabolites, MAPK pathway, glutathione metabolism, and photosynthetic carbon fixation.

In our results, GO enrichment analysis revealed that the upregulated DEGs induced by exogenous GABA in M.9-T337 under well-watered conditions were enriched in the terms of JA, SA, ABA, amino acid, and organic acid (Fig. S5B). As important regulators, plant hormones are involved in sensing and transmitting various environmental signals and play an important role in plant response to abiotic stresses. Previous studies found that SA, JA, and ABA play important roles in plant response to drought stress (Wang et al., 2021e; Ahmad et al., 2021; Raghavendra et al., 2010). Exogenous GABA can cause an increase in GABA content in plants (Li et al., 2021a). Organic acids are important intermediates in the metabolism of substances in plants, which play an important role in the adaptation of plants to various stresses (Chen et al., 2009). Therefore, GABA pretreatment may influence the plant processes related to JA, SA, ABA, amino acid, and organic acid, which in turn affects plant drought resistance.

In summary, we investigated the effects of exogenous GABA on apple fruit quality during fruit growth and drought tolerance. Our results revealed that exogenous GABA could improve fruit quality, including increased sugar-to-acid ratio, flesh firmness, pericarp malleability, and GABA content, as well as reduced fruit acidity. Moreover, 1 mM exogenous GABA improved drought tolerance to both long- and short-term drought stress. Together, the findings indicated that exogenous GABA can improve fruit quality and enhance drought tolerance in apple, which is of great significance for GABA application in the apple production and cultivation.

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CRediT authorship contribution statement

Pengda Cheng: Formal analysis, Investigation, Writing – original draft. **Qianyu Yue:** Formal analysis. **Yutian Zhang:** Investigation. **Shuang Zhao:** Formal analysis. **Abid Khan:** Writing – original draft.

Xinyue Yang: Investigation. **Jieqiang He:** Formal analysis. **Shichong Wang:** Formal analysis. **Wenjun Shen:** Formal analysis. **Qian Qian:** Formal analysis. **Wanshan Du:** Formal analysis. **Fengwang Ma:** Formal analysis. **Dehui Zhang:** Writing – original draft, Writing – review & editing. **Qingmei Guan:** Project administration, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The RNA-seq data have been deposited to the NCBI with the dataset identifier PRJNA847788.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jplph.2022.153890>.

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